

SPARE PART CATALOG

12V4000M70/526103698



Power. Passion. Partnership.

Printed in Germany

© 2014 Copyright MTU Friedrichshafen GmbH

This Publication is protected by copyright and may not be used in any way whether in whole or in part without the prior written permission of MTU Friedrichshafen GmbH. This restriction also applies to copyright, distribution, translation, microfilming and storage or processing on electronic systems including data bases and online services.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments **Business Portal Online Print 06-06-2023**

Preface

This catalog is the basic reference document for ordering spare parts. Illustrations are provided for extensive assembly operations to assist trained personal in conducting such operations but do not claim to be complete. For better overview, even product components that are not available as spare parts are shown.

The individual components were taken from the assembly list.

Catalog structure

The catalog is divided into Function Groups (FG) and Function Group Segments (FGS) in ascending order.

The side contain the following information:

- **MODEL:** (Product Type)
- **SERIAL NO.:**
- **FG/FGS DESIGNATION**
- **FG/FGS NO.:**
- **BOM No.:**

Explanation of columns

- **ITEM:** Provides a cross-reference between illustration and text section.
- **PART NO.:** This is the valid number for ordering the part in question.
- **DESIGNATION:** The precise part name.
- **MODEL/TYPE** is an additional specification.
- **QTY.: (Quantity in Unit):** Indicates the quantity of this part required for the case in question. Quantities not stated are available on request.
- **UNIT:** The 3-character codes indicate: PC = Piece, KG= Kilogram, M= Meter, ROL= Roll, PAC= Pack
- **WEIGHT**
- **FN.:** (Footnote) Provides additional parts information. For clarification of footnote symbols, see Page 2.
- **DIMENSIONS:** This column contains standard-part dimensions or manufacturer designations for purchased parts.
- **STANDARD**

Amendments service

Part No. changes and amendments which occur after publication of the catalog are not published. We therefore recommend that users make a record of amendments in their catalogs. This information is provided, for instance, in order confirmations and/or on spare parts labels.

Footnotes: Key

AB	Only available complete. Repair by manufacturer only.
AC	Not scheduled as a spare part. Availability must be checked.
ACN	Not scheduled as a spare part. Availability must be checked. Component only available complete.
EM	Emission-relevant part. In the case of all marine engines with IMO certification it should be noted that the identification numbers of the emission-relevant parts on the engine must match the details given in the technical file. When ordering, quote EM code with order no. E.g: 535 030 04 17 EM.
EMN	Emission-relevant parts. Component only available complete.
KD	Group balancing required
KE	Individual part; not scheduled as a spare part.
KF	Adjusted spare part.
KFN	Adjusted spare part. Component only available complete.
M	Spares-related components or parts with different repair sizes. Dimensions are listed in the Tolerances and Wear Limits List.
MN	Spares-related components or parts with different repair sizes. Components only available complete.
MUB	Spares-related components or Parts with different repair sizes. Available only as complete set.
NN	Component is not broken down into further sub-components.
P	List of bought-in components
SB	Group of parts which are welded or brazed together
UA	Indicate dimensions when ordering
UB	Only supplied complete or as a complete set.

Ordering Procedure

The following information is essential for spare-part order processing:

- Engine Model
- Serial number (see engine name plate)
- FG/FGS
- Item
- Part No.
- Designation
- Model/Type
- Quantity
- Page

When ordering replacement parts for ancillary equipment such as injection pumps, fuel pumps, etc., give both model and part number of the part.

When ordering a calibrated part (marked by "KF", "KFN" in the "FN" column), type the following additional information in the **Comments** field:

If the calibration No. is known (see part's nameplate):

Add the calibration No., preceded by an oblique stroke, as an extension to the part number ("/xxxxx")

e.g.: 0025362810/E1234 Here, the extension "E1234" would be the part's calibration number on the nameplate.

If the calibration No. is not known:

Give a detailed description of the required part's installation location by quoting the complete Device Description code (e.g. "= 37.1.1 + A022 - F003") plus the part number extension "/77" (e.g. 5005362810/77).

In this case, the calibrated spare part will be supplied under a new part number.

The Device Description is found on the nameplate at the part's installation location, or in the corresponding Wiring Diagram.

When ordering a non-calibrated adjustable part (marked by "KF", "KFN" in the "FN" column) which should not be calibrated by the factory: Add extension "/76" to the part number (e.g. 0025362810/76).

Phone orders should be confirmed in writing to prevent errors.

Ordering Example:

Engine model	Serial number	FG/FGS	ITEM	Part No.	Designation	Quantity	Page
16V4000R41	527101836	031 88	5	5270300701	Crankshaft	1	33
		083 05	10	0020922801	Easy-Cange Filter	2	92
Blue Line	V4000M Waterjet	517 00	75	5185301593	Control Console	1	53

Factory repairs

Parts shipped to MTU for repair must be marked with the relevant engine serial number (tag or label, for example).

Exchange components (Reman)

Exchange components are parts overhauled by the manufacturer. They have the same service life and quality characteristics as new parts.

Components which are included in the Reman exchange program are marked by the following letters in front of the item number:

For instance: **E 524 010 3520 = New component in exchange program**

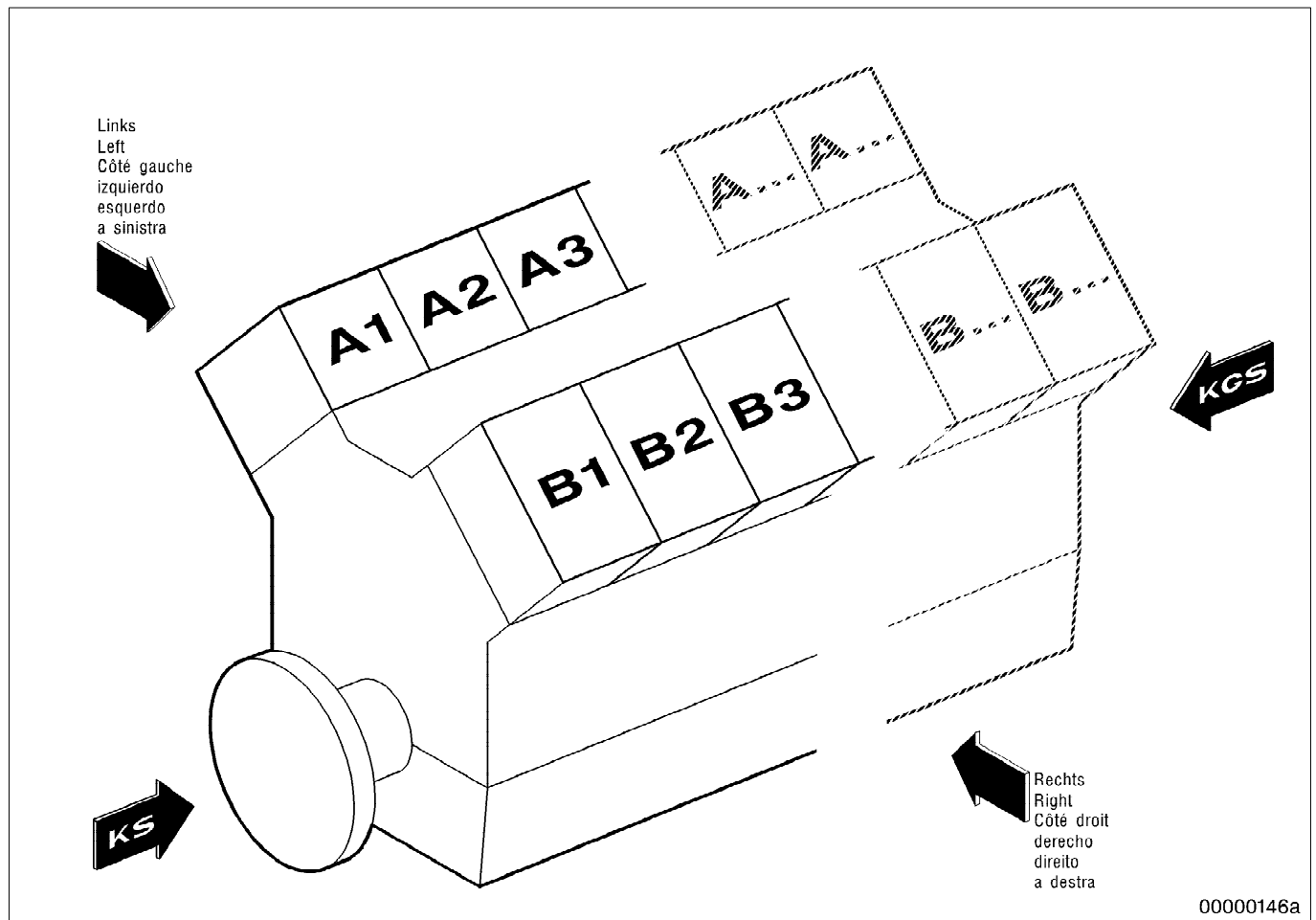
R 524 010 3520 = Reman component in exchange program

Exchange procedure

With the introduction of exchange components, the so-called core deposit will also be charged for new parts. The core deposit will be indicated separately on the invoice and the respective value will be credited upon return of reusable components.

Further information upon request.

Engine side and cylinder designations



Engine sides / banks are always designated as viewed from the driving end (KS).

The left bank of cylinders is designated "A" and the right bank "B" (according to DIN ISO 1204).

The cylinders of each bank are numbered consecutively from the driving end, starting with No. 1.

Other engine components are also numbered consecutively from the driving end, starting with No. 1.

Designations and abbreviations

Driving end = KS

Free end = KGS

Left-hand side

Right-hand side

Top

Bottom

Turbocharger = ATL

FG FG DESIGNATION**004 LABELS AND AUXILIARY MATERIALS**

011 CRANKCASE

014 OIL PAN

015 ENGINE MOUNTING

016 END HOUSING, DRIVING END AND FREE END

018 CRANKCASE BREATHER

024 GEARCASE, AUX PTO END

031 CRANKSHAFT WITH BEARINGS

032 CRANKSHAFT, PTO ON DRIVING END

035 CRANKSHAFT, PTO ON AUX PTO END

036 TRANSPORT LOCK/LOCK FOR CRANKSHAFT

037 PISTON

038 CONNECTING ROD

041 CYLINDER HEAD

051 CAMSHAFT

055 VALVE DRIVE

073 INJECTION PUMP / HP PUMP

075 FUEL INJECTOR

077 INJECTION LINE / HP FUEL LINE

081 FUEL PIPEWORK BEFORE DELIVERY PUMP

083 FUEL FILTER

086 LEAK-OFF FUEL LINE / FUEL RETURN

101 EXHAUST TURBOCHARGER, GENERAL OR H.P. TURBOCHARGER

111 CHARGE AIR COOLING, GENERAL, LEFT SIDE

121 AIR FILTER

123 AIR SYSTEM

124 AIR SUPPLY TO CYLINDERS

126 AIR TRANSFER, LEFT - RIGHT SIDE, DRIVING END

128 CONTROL BLOCK FOR AIR AND EXHAUST FLOW CONTROL FLAPS

142 CONSTANT-PRESSURE EXHAUST MANIFOLD

143 CARRIER HOUSING

144 EXHAUST FLOW CONTROL FLAP WITH CONTROLS

146 EXHAUST DUCTING AFTER ENGINE (TO END OF EXH. AFTERTREATMENT)

172 ELECTRIC STARTING

181 LUBE OIL PUMP AND DRIVE

183 OIL FILTRATION/COOLING

185 TURBOCHARGER OIL SUPPLY

188 CONNECTING HARDWARE FOR ENGINE OIL SUPPLY

200 COOLANT SYSTEM

202 COOLANT CIRCUIT, GENERAL, HT CIRCUIT

204 HEAT EXCHANGER PLATES

206 COOLANT PIPEWORK WITH THERMOSTAT

207 RAW WATER PUMP WITH CONNECTIONS

208 COOLANT EXPANSION TANK

213 GENERATOR

231 ENGINE MOUNTING

FG FG DESIGNATION

- 250 MAIN AND AUXILIARY PTO SYSTEMS (COUPLINGS)
- 251 STEEL-SPRING COUPLING
- 293 OIL PUMP
- 295 FUEL PUMPS
- 301 GENERAL ACCESSORIES
- 303 PNEUMATIC ACCESSORIES
- 362 COMPRESSED AIR SUPPLY SYSTEM
- 365 EXHAUST SYSTEM (AFTER ENGINE / AFTER EGA)
- 368 RAW WATER SUPPLY AND DISCHARGE SYSTEM
- 507 WIRING, GENERAL, FOR ENGINE/GEARBOX/PLANT
- 508 ACCESSORIES FOR ENGINE GOVERNOR (ELECTRONIC) CONTROL SYSTEM
- 510 MONITORING EQUIPMENT
- 511 ALARM SYSTEM
- 516 RECORDING SYSTEM (FAILURES/MANOEUVRES/EVENTS PRINTING)
- 517 EMERGENCY INSTRUMENTATION (CONTROL STAND)
- 526 REMOTE CONTROL